AnnRakshak AI - System Implementation & UI Architecture Guide

Operational Documentation, 3 Selectable UI Themes, Component Usage Matrix & Audit Report

Surplus Food Rescue • Kitchen Waste Audit • Predictive Demand Planning • Gemma 4 AI

Document Version: 2.0 (Audited Release)	**Target Locality:** Prayagraj, India (IST)	**Generated:** 2026-10-19
Tech Stack: Streamlit, SQLite (WAL), Plotly, google-genai	**AI Model:** Gemma 4 (gemma-4-26b-a4b-it / 31b)	**Supported Languages:** English (en), Hindi (hi)

1. Executive Summary & Problem Statement

Commercial banquets, wedding venues, corporate events, and hotel kitchens frequently face unpredicted service variance. According to the *UNEP Food Waste Index Report (2024)* and Indian hospitality catering studies, approximately 10% to 20% of prepared banquet food is routinely discarded due to RSVP dropouts, adverse weather, or banquet closure timing. Simultaneously, night shelters and vulnerable communities in urban areas experience persistent meal deficits. **AnnRakshak AI** bridges this gap through four core capabilities:

- **Surplus Food Rescue & Coordination:** Canonical 5-step milestone lifecycle (Available -> Requested -> Confirmed -> Collected -> Delivered) connecting donors with verified NGOs.
- **Kitchen Food Waste Auditing:** Logging portions prepared vs consumed, calculating edible surplus vs inedible discards, and quantifying financial loss (INR).
- **Predictive Demand Planning:** Mathematical estimator using historical consumption ratios, attendees, and clamped buffers (0-50%) to prevent over-preparation before cooking starts.
- **Gemma 4 AI Insights:** Synthesizing kitchen audits and active surplus listings using Google Gen AI SDK to recommend waste reduction strategies and dispatch priorities.

2. Full Portal Navigation Map

The application provides three dedicated portal perspectives secured by role-based PIN authentication (Donor PIN 1234, NGO PIN 5678, Admin PIN 9999). Each portal features numbered navigation sections:

Portal Perspective	Section Number & Name	Key Functional Purpose
Donor Portal (Food Providers / Caterers)	1. Overview 2. Donate Food 3. My Donations 4. NGO Pickup Requests 5. Food Waste Tracker 6. Demand Prediction 7. Gemma 4 AI Insights 8. Impact Summary	Operations KPIs, surplus food publishing, multi-stage coordination, kitchen waste audits, predictive demand calculations, AI waste advisory, and CSV exports.

Portal Perspective	Section Number & Name	Key Functional Purpose
NGO Portal (Food Recovery Organizations)	1. Overview 2. Available Food Donations 3. Search & Filters 4. Request Pickup 5. My Pickup Requests 6. Collection & Delivery Status 7. Impact Summary	Surplus discovery feeds, urgency countdown tracking, partial portion claims, driver coordination, collection/delivery milestone progression, and impact reporting.
Admin Portal (System Administrators)	1. NGO Verifications 2. System Audit Trail 3. Reset Database	Reviewing and granting verified standing to NGO partners, monitoring tamper-proof status transition logs, and non-destructive demo database management.

3. Complete Catalog of the 7 Interactive Plotly Charts

All visual analytics are rendered using Plotly interactive charts and automatically synchronized to the active theme palette via apply_plot_theme():

Chart # & Title	Portal & Section Location	Chart Type & Visual Metrics Tracked
Chart 1: Donation Status Distribution	Donor Portal: Section 1 (Overview)	Donut Chart showing proportion of listings in Available, Requested, Confirmed, Collected, Delivered, and Cancelled.
Chart 2: Portions Offered vs Collected vs Delivered	Donor Portal: Section 1 (Overview)	Grouped Bar Chart comparing total meal portions listed against portions loaded into vans and verified delivered.
Chart 3: Recent Consumption vs Unconsumed Leftovers	Donor Portal: Section 5 (Waste Tracker)	Stacked Bar Chart showing chronological service logs with consumed meals stacked against unconsumed leftovers.
Chart 4: Financial Discard Loss by Meal Category	Donor Portal: Section 5 (Waste Tracker)	Pie / Donut Chart visualizing aggregate monetary loss (INR) distributed across Dinner, Lunch, Breakfast, and Buffets.
Chart 5: RSVP vs Estimated Net Demand vs Prep	Donor Portal: Section 6 (Demand Predictor)	Grouped Bar Chart contrasting raw attendee RSVP count against historical arrival demand and recommended batch size.
Chart 6: Surplus Meal Portions by Area / Locality	NGO Portal: Section 1 (Overview)	Horizontal Bar Chart displaying available rescue portions grouped by geographic banquet clusters in Prayagraj.
Chart 7: NGO Pickup Request Status Breakdown	NGO Portal: Section 1 (Overview)	Donut Chart tracking active recovery missions across Requested, Confirmed, Collected, and Delivered states.

4. Selectable UI Setups & Contrast Audit

To provide accessibility across indoor kitchen offices, outdoor lawns, and evening transport routes, AnnRakshak AI features three selectable UI setups. All color pairings have been validated via automated contrast script (check_contrast.py):

Theme Name	Palette & Tone	Target Operational Environment	Audited Contrast Ratio
AnnRakshak Nature ***(Warm Forest & Cream)***	• Canvas: #FBF9F4 • Cards: #FFFFFF • Primary: #1B4332 • Accent: #A23B20 • Text: #0F172A	Daytime kitchen management, banquet reception desks, and administrative logging.	**16.97:1** (Body on bg) **6.60:1** (Accent on white) WCAG AA > 4.5:1 [PASS]
Midnight Pro ***(Dark Obsidian Mode)***	• Canvas: #0B0F19 • Cards: #151E2E • Primary: #10B981 • Accent: #38BDF8 • Text: #F8FAFC	Night-time recovery vans, low-light shelter distribution, and evening catering audits.	**18.30:1** (Text on dark) **6.59:1** (Mint on surface) WCAG AA > 4.5:1 [PASS]

Theme Name	Palette & Tone	Target Operational Environment	Audited Contrast Ratio
High-Contrast Daylight ***(Executive Navy & Gold)***	• Canvas: #FFFFFF • Cards: #FFFFFF • Primary: #0A192F • Accent: #854D0E • Text: #000000	High-glare outdoor wedding lawns, dockside logistics, and projector presentations.	**21.00:1** (Black on white) **6.85:1** (Gold on white) WCAG AAA > 7:1 [PASS]

5. System Architecture & Core Module Matrix

AnnRakshak AI utilizes a centralized SQLite database operating in WAL (Write-Ahead Logging) mode with busy_timeout=5000ms and BEGIN IMMEDIATE locks for serialized writer access:

Module / File	Core Responsibilities & Capabilities	Key Methods & Architectural Features
database.py	Initializes tables: donations, pickup_requests, food_waste_records, ngos, status_history. Seeds fictional Prayagraj records non-destructively.	init_db(), get_connection(). WAL mode, foreign keys, IST (Asia/Kolkata) timezone-aware datetimes.
workflow_manager.py	Implements canonical status enum, partial requests, atomic portion release on Rejection/Cancel, auto-expiry, and CSV formula injection protection.	WorkflowStatus enum, create_donation(), request_pickup(), update_request_status(), check_and_expire_donations().
demand_predictor.py	Deterministic baseline estimator computing food preparation targets from historical consumption ratios, attendees, and clamped buffers (0-50%).	calculate_demand_estimate(). Handles 0/negative attendees, clamped cushions, and cold-start hospitality fallback ratios.
ai_insights.py	Google Gemma 4 AI integration via google-genai SDK. PII phone/email scrubber, model verification check, timeout retries, and bilingual output.	analyze_food_waste_with_gemma(), prioritize_donations_with_gemma(), verify_gemma_model().
i18n.py	Internationalization engine supporting English (en) and Hindi (hi). Canonical status translations, PIN auth, and environmental impact keys.	t(key, lang, **kwargs). Dictionary-based string lookup with parameterized token interpolation.
ui_styles.py	Modular styling engine generating WCAG AA compliant CSS, responsive listing cards, status badges, metric boxes, and 5-stage visual milestone steppers.	get_theme_css(), render_workflow_stepper(), render_donor_listing_card(), render_ngo_available_card().
check_contrast.py	Standalone accessibility script calculating WCAG 2.1 relative luminance and contrast ratios for all three application themes.	calculate_contrast_ratio(), check_all_themes(). Verifies compliance (ratio >= 4.5:1 for body text).
test_workflow.py	Automated test suite executing 9 isolated verification tests (Tests A-I) against a temporary SQLite database (never touches annrakshak.db).	TestAnnRakshakAIAudit. Tests partial claims, portion release, 10-thread concurrency lock, and CSV escaping.

6. Automated Verification Test Suite Results (Tests A through I)

All test runs execute against a temporary SQLite database file using tempfile.NamedTemporaryFile, guaranteeing complete demo database isolation. All 9 tests passed:

Test Identifier	Tested Functional Dimension	Assertion & Audit Verification Result
Test A	Wedding Donation & Food Safety Window	Asserts valid creation with IST time; verifies food prepared >6 hours ago is blocked. **PASSED [OK]**
Test B	Partial Requests & Portion Hold	Asserts partial claim (40/100 portions) leaves 60 portions Available for other NGOs. **PASSED [OK]**
Test C	Milestones & Environmental Impact	Progresses to Delivered; verifies 17.5 kg food rescued and 43.75 kg CO2 avoided. **PASSED [OK]**
Test D	Atomic Portion Release on Reject/Cancel	Asserts rejected/cancelled claims immediately return portions to the Available pool. **PASSED [OK]**

Test Identifier	Tested Functional Dimension	Assertion & Audit Verification Result
Test E	Auto-Expiry of Past-Deadline Listings	Asserts past-deadline listings and unconfirmed requests are auto-marked Expired. **PASSED [OK]**
Test F	Concurrency & No Double-Booking	10 concurrent threads attempt to claim 40 portions from a 50-portion pool; exactly 1 succeeds, 9 fail. **PASSED [OK]**
Test G	Demand Predictor Edge Cases & Clamping	Asserts 0/negative attendees raise ValueError; buffer clamped strictly between 0-50%. **PASSED [OK]**
Test H	Gemma Mocking, PII Redaction & Fallback	Asserts donor phone/email scrubbed from prompt; verifies deterministic Hindi fallback. **PASSED [OK]**
Test I	CSV Formula Injection Escaping	Asserts cells starting with =, +, -, @ are safely escaped with leading single quote. **PASSED [OK]**

System Status: Fully Audited & Operational **Live Server:** http://localhost:8501 **Database:** SQLite (annrakshak.db, WAL)